

Euclidean and Non-Euclidean Geometries

A learner-centered approach to axiomatic thinking for everyone, including “poets¹.”

Charles A. Coppin

Lamar University

¹The word “poets” endearingly refers to all undergraduates majoring in the humanities, social sciences, or the fine arts.

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Preface

It was a time, long ago, when the author was young, naïve, and, prematurely, chair of a department of mathematics when he recommended that a *proof course*², a shortened version of axiomatic geometry for mathematics majors, be required of all “poets”³. The alternatives, then, as now, were to teach liberal arts students, a hodgepodge of “interesting” mathematical topics or something “useful” such as college algebra or statistics.

I believe that such a mathematical stew does not engage the “poet” in the essence of mathematics but only perpetuates the usual stereotypes. Mathematics has just as much a claim on truth as the humanities. There is a deep resonance between mathematics and the humanities. You do not see this in the typical mathematics courses for liberal arts students. Potentially, there is a better connection between mathematics and the humanities than between the experimental sciences and the humanities. In fact, it was my colleagues from the humanities that encouraged me the most, not my colleagues from the experimental sciences who, if they spoke to me about this matter, would advocate some “useful, check-book” mathematics course. Today, removed in time and place, I am not so sure that I would have the courage to implement a course wherein almost all “poets” do proofs on their own. Oh, is not the naïveté of youth refreshing! As it turns out, the course has been quite successful and very exciting for both students and teachers. Colleagues at private and public universities who have been teaching this course have immensely encouraged me.

Although most of these notes are of my own creation, actually, as a response to my own students’ reactions to the notes, I have referenced several sources over the years. Among these are the following: *Euclidean and Non-Euclidean Geometries* by Marvin Jay Greenberg, *Foundations of Euclidean and Non-Euclidean Geometry* by E. B. Golos, *Synthetic Geometry* by R.F. Jolly, and a set of unpublished notes by James N. Younglove. The format of these notes follows somewhat the unique format of *Groups: A Pathway to Geometry* by R. P. Burn.

²A *proof course* is a mathematics course with the primary emphasis of teaching students to do proofs.

³By using the word “poet”, I mean liberal arts majors; that is, majors in the humanities and the arts.

Acknowledgments

Special thanks go to all my many students over the decades who have learned from these notes and who have helped to improve them.

Finally, Alaine Fay, my wife of many years, has heard me talk about this course, has heard me talk about the students and has inspired my teaching and the design of my courses over the years.

To the Student

You may be a liberal arts major, a fine arts major, an engineering major, a science major, or even a major in mathematics, one of the most beautiful subjects I know. However, you may be wondering why such a course as this one would be of any value to you. As I state in the subtitle, these notes are for each and every one of you. Allow me to explain.

The best academic representation of authentic mathematics are courses where deductive reasoning is dominant; proofs are strongly emphasized. How do we know mathematical truth? Truth in the experimental sciences is by way of the scientific method or induction; however, mathematics is very different from the standard mathematics courses normally required of all majors. Mathematics at its heart relies strictly on deduction as its means of validating truth. This means proofs! Speak with any mathematician, and he or she will speak of theorems, conjectures, and proofs. Starting with Euclid's *Elements* to this present day, real mathematics is the creation of proofs. Why do I speak of courses rather than merely written or recorded mathematics? Mathematics exists only in the present. A screenplay is not a play until actors actually perform the written work on a stage. When the curtain falls, the play is over. It only truly exists in the here and now. The same is true of mathematics. It is in class where budding mathematicians (students) or mature mathematicians perform their mathematics. In class, original proofs are presented, conjectures are made, examples and counter examples are given, the imagination is engaged, excitement abounds and, on a good day, pure poetry flows. This is at the highest level; this is authentic mathematics and proof is at its core!

Mathematics is truly a discipline for *every* educated person. Over a millennium, quoting Felix Browder, mathematics came to be identified with the great structure of deductive geometry as embodied in Euclid's *Elements* and the writings of Archimedes. The mathematics courses we teach today, especially, to our poets are quite utilitarian or have a focus on an objective outside of mathematics itself not on the essence of mathematics.

This course can be challenging or even frustrating at times; however, this is part of the intellectual pursuit. If you reflect on anything worthwhile you have accomplished such as training for a marathon, learning a musical instrument, or learning to be a ballet dancer, you will remember there was pain and frustration but weren't the rewards great? I hope this will be true here. The culture of the class is decidedly different from your previous mathematics courses but may be more like your training in athletics, the band hall, or your dance class. You learned by doing didn't you? The same is true here. Forget about your previous mathematics courses; immerse yourself in this world of authentic mathematics. Listen to and trust your teacher. He or she is your coach in one of the most enjoyable and rewarding of academic activities and is your guide on one of the best intellectual journeys you will ever undertake.

Chapter 1

Boot Camp

This chapter is intended to be covered very quickly. We do not intend to actually teach logic nor develop a deep understanding of set theory. The language section serves the purpose of putting certain vocabulary in place. The set theory section is merely there to give students an opportunity to learn a little about proofs before hitting the main event, the chapter entitled *Neutral Geometry*. The section on *Axiom Systems* in this chapter should be covered carefully and in some detail.

1.1 Language

The concepts found in doing proofs could be familiar to the reader but some of the words may be new. We list those below and elaborate on their deeper meaning and usage.

We state two rules of logic, which are the cornerstone not only of proof in mathematics but of all intelligent human discourse.

Law of Excluded Middle. Each statement is true or false.

Law of Contradiction. No statement is both true and false.

These two laws come to us from Aristotle. See how these are applied in Examples 2 and 3 below. But, first, we learn some basic vocabulary of logic.

Consider the logical connectives *and* and *or*. We say that the statement *S and T* is true if and only if both *S* and *T* are true where *S* and *T* are statements. The statement *S or T* is true if and only if at least one of *S* and *T* is true.

An **implication** is a statement of the form : *if P, then Q*. *If P, then Q* is stated symbolically as $P \Rightarrow Q$ for those who are so led. Under what circumstances is the statement *if P, then Q* true? Or, a better way, is to ask the question when is *if P, then Q* not true? When does someone lie using implication? It is when he or she says “I shall do such and such” and then does not do “such and such.” That is, the implication *if P, then Q* is false only when *P* is true and *Q* is false. Thus, *if P, then Q* is true only in case one of the following three statements hold:

P and *Q* are both true,
P and *Q* are both false,
P is false and *Q* is true.

This is summarized in the following table:

P	Q	$\text{if } P, \text{ then } Q$
True	True	True
True	False	False
False	False	True
False	True	True

P is known as the **hypothesis** and Q is known as the **conclusion** of the statement *if P , then Q* . If P and Q denote statements, a **deduction** is as follows:

If the statement *if P , then Q* is true and the statement P is true, then the statement Q is true.

Symbolically, in the world of formal logic, deductions are written in the following form

$$\frac{\begin{array}{c} \text{if } P, \text{ then } Q \\ P \end{array}}{Q}$$

You use deductions in every day life. The following is an example of a deduction:

If Hobbes awakens before Calvin, he will prepare breakfast.
Hobbes awakens before Calvin.
Therefore, Hobbes prepares breakfast.

The **negation** of a statement is rather straightforward when the statement is a simple declarative statement with no quantifiers such as “each”, “every”, “there is”, and “there exists.” Thus, the negation of S = “grass is green” is $\neg S$ = “grass is not green.” Another name for negation used by teachers of an earlier generation is “bare denial.” This terminology is better in that it connotes the real idea behind negation, which is that negations are minimally true or just barely true. In other words, when one wants a negation of a statement, one wants a statement that is complementary to the original statement. For example, let S = “every cow in the field is brown.” One could say that $\neg S$ = “not every cow in the field is brown” is a negation or bare denial of S , and this is correct. However, as you will see, this form of negation is not useful. We desire a negation that uses the most positive language possible. When negating a statement, move the “no” or “not” as far as possible to the right in the sentence that is to be the negation. Therefore, a better choice of negation is $\neg S$ = “there is some cow in the field that is not brown.” Note that $\neg S$ is complementary to S . Choosing “every cow in the field is not brown” is not correct. It goes too far in negation.

There is one thing that a serious student of mathematics must learn to do. He or she must learn how to determine a bare denial of a statement. Try your hand at developing bare denials of the statements below but cover up the column on the right so as not to see the correct answer. Then check yourself against the answers in the right column.

<u>Statement</u>	<u>Bare Denial</u>
Tom ate a horse.	Tom did not eat a horse.
Each cow has two heads.	Some cow does not have two heads.
Some roof of some house has a hole.	No house has a hole in its roof.
I shall pass or I shall graduate.	I shall not pass and I shall not graduate

Now, work the following exercise.

Exercise 1. Develop a bare denial or negation of the following statements:

- (a) All little green men have horns.
- (b) Every math major is smart.
- (c) I shall major in biology or physics.

We, now, consider other concepts and words of language.

- A **theorem** is a statement that is always true. For example, the following is a theorem:

The product of two non-zero real numbers is a non-zero real number.

- A **contradiction** is a statement that is always false. The best form of contradiction is that S and not S are both true where S is a statement. For example, the following statement is a contradiction:

The president is a whig and the president is not a whig.

- A **proof** is a collection of statements each step of which must be supported by appeal to the hypothesis, definition, axiom, a previous theorem, or by deduction. See Examples 1, 2, and 3.
- A **direct proof** is a list of statements where the first item in the list is the hypothesis and the last statement is the conclusion of the theorem. The other statements are supported. See Example 1.
- An **indirect proof** of a statement S begins with a negation of the statement to be proved. Thus, we begin the indirect proof with $\neg S$. From this point, there ensues a sequence of statements, each one supported by appeal to definition, axiom, a previous theorem, or by deduction, just as in a direct proof. The last statement in the sequence, say Q , is a contradiction; that is, Q is not true. However, by deduction, since $\neg S \rightarrow Q$ is true and $\neg S$ is true, then Q is true, which is impossible. When you get to the point that a contradiction results and you know all steps of the proof are valid, then you know that the fault lies with the fact that $\neg S$ was assumed to be true but is actually false. Thus, S must be true.

Example 1. Consider the following true statement (called a theorem) and a direct proof of the statement. We shall assume the following axiom:

Commutative Axiom. For each number x and each number y , $x + y = y + x$.

Theorem 1. If $a + y = y$ for each number y and $b + y = y$ for each number y , then $a = b$.

Proof. Suppose $a + y = y$ for each number y , b is a number, and $a + b = b$. Since $b + y = y$ for each number y and a is a number, then $b + a = a$. By the Commutative Axiom, $a + b = b + a$. Therefore, since $a = b + a = a + b = b$, then $a = b$. $\square$

We see in Example 2 below how one proves statements to be false. We do this by finding a counter example, a case that proves a particular general statement to be false.

Example 2. Consider the statement: If each of a and b is a number, then $a + b = a \cdot b$.

Solution. This statement is true for some values of a and b , for example $a = 0$ and $b = 0$; however, this is irrelevant. The important thing is that it is not true for $a = b = 1$. On substitution, we obtain $2 = 1$, a contradiction. This one case is sufficient to prove that the statement is not universally true.

Example 3. This example demonstrates proof by contradiction or indirect proof.

Theorem 2. If $w \geq 0$ and $w < c$ for each positive number c , then $w = 0$ or, equivalently, each non-negative number that is less than each positive number is zero.

Proof. Assume that the statement is not true. Then, by the Law of Excluded Middle, its negation must be true. Thus, there must be some $w > 0$ such that $w < c$ for each number c . Since $w > 0$, we conclude that $w < w$. This is a contradiction. Therefore, the original statement is true (a theorem). $\square$

1.2 Set Theory

We shall not attempt a formal definition of “set”, but suffice it to say that a set is a group or collection of objects. The symbol $\emptyset$ denotes the set consisting of no elements and is known as the empty set (If there is such a thing!).

Example 4. The following are examples of sets: the set of numbers, the set of colleges in the United States, the set of planets in the solar system, the set of students in Math 3321, and the set of unicorns in the Bronx zoo (Is there such a thing?).

Moreover, the words, “collection” and “family”, can be used in place of the word “set.” Sometimes, these words can be used to indicate an hierarchical relationship among sets. For instance, one could speak of a collection of sets and a family of collections.

The following are different ways of describing a set.

- (a) Listing. For example, $\{1, 2, 3\}$ is the set consisting of the numbers 1, 2, and 3.
- (b) Set Criteria. For example, $\{x : x \text{ is a number and } x > 0\}$ is the set of all positive numbers.

Notation 1. When we want to say that x is an element of the set M or x belongs to M , we may use the symbols $x \in M$. Of course, $x \notin M$ means x does not belong to M .

Fact: Suppose $M = \{x : P(x)\}$ where $P(x)$ is a sentence about x (such as “ x is a number and $x > 0$ ” above). When we see $z \in M$, then we know that $P(z)$ is true. Also, when $P(z)$ is true, then we automatically know that $z \in M$. For example, suppose $M = \{x : x \text{ is a planet in the solar system}\}$ where $P(x) = \text{“}x \text{ is a planet in the solar system.”}$ Since $P(\text{Mars})$ is true, then $\text{Mars} \in M$. Moreover, if $z \in M$, then we know that $P(z)$ is true; therefore, z must be a planet in the solar system. In summary, $(z \in M)$ is equivalent to $(P(z) \text{ is true})$ where $M = \{x : P(x)\}$. Remember this last fact as it will be most useful throughout the course.

Definition 1. Suppose each of A and B is a set. The phrase “ A is a subset of B ” means that if x belongs to A , then x belongs to B or, using shorthand, $x \in A$ implies $x \in B$. We write “ A is a subset of B ” as $A \subseteq B$.

Example 5. Consider the set $\{1, 2\}$. The collection of all subsets of $\{1, 2\}$ is

$$\{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$$

Theorem 3. If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

Definition 2. Suppose each of A and B is a set. A is said to be equal to B (written $A = B$) if and only if $A \subseteq B$ and $B \subseteq A$.

Definition 3. Suppose each of A and B is a set.

- (a) The difference of A and B (written $A \setminus B$) is the set of all x belonging to A and not to B ; using other symbols, $A \setminus B = \{x : x \in A \text{ and } x \notin B\}$.
- (b) The intersection of A and B (written $A \cap B$) is the set of all x belonging to both A and B ; using other symbols, $A \cap B = \{x : x \in A \text{ and } x \in B\}$.
- (c) The union of A and B (written $A \cup B$) is the set of all x belonging to A or to B ; using other symbols, $A \cup B = \{x : x \in A \text{ or } x \in B\}$.

Historical Note. In the “old” days (first half of the twentieth century), $A \cap B$ was written as $A \cdot B$ and known as the intersection (common part) of A and B . Also, $A \cup B$ was denoted as $A + B$ and known as the sum of A and B .

Theorem 4. Suppose each of A and B is a set. Then

- (a) $A \subseteq A \cup B$
- (b) $A \cap B \subseteq A$
- (c) $A \subseteq B$ if and only if $A \cup B = B$

Optional Material

Theorem 5. Suppose each of A , B and C is a set. Then

- (a) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- (b) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- (c) $A \subseteq B$ if and only if $A \cap B = A$.

Definition 4. Suppose U is a set and $B \subseteq U$. Then, by the complement of B with respect to U (written B^c and read “ B complement”) we mean $U \setminus B$. Notice that the meaning of B^c is dependent on the set U .

Example 6. If $U = \{1, 2, 3, \dots\}$, then $\{1, 3, 5, \dots\}^c = \{2, 4, 6, \dots\}$. However, $\{-2, 0, 2, 4, 6, \dots\}^c$ would have no meaning unless we change our definition of U .

Theorem 6. Suppose U is a set and each of A and B is a subset of U . Then, forming complements with respect to U ,

- (a) $(A^c)^c = A$
- (b) $A \cup A^c = U$
- (c) $A \subseteq B$ if and only if $B^c \subseteq A^c$
- (d) $(A \cup B)^c = A^c \cap B^c$
- (e) $(A \cap B)^c = A^c \cup B^c$

Conjecture 7. Suppose each of a , b , c , and d is an object. Then $\{a, b\} = \{c, d\}$ if and only if $a = c$ and $b = d$.

Definition 5. Suppose G is a collection of sets.

- (a) The intersection of G (written $\cap G$) is the set of all x belonging to each member of G ; using other symbols, $\cap G = \{x : x \in g \text{ for each } g \in G\}$.
- (b) The union of G (written $\cup G$) is the set of all x belonging to at least one member of G ; using other symbols, $\cup G = \{x : x \in g \text{ for some } g \in G\}$.

Theorem 8. Suppose U is a set and G is a collection of subsets of U . Then, forming complements with respect to U ,

- (a) $(\cup G)^c = \cap \{g^c : g \in G\}$
- (b) $(\cap G)^c = \cup \{g^c : g \in G\}$

1.3 Axiom Systems

An **axiom system** has two components:

- (a) A set of special words that we officially assume have no meaning. The words are sometimes called *undefined terms*, *primitives* or *technical terms*.
- (b) A set of statements called *axioms* that we assume to be true concerning the undefined terms.

One of our favorite examples of an axiom system is as follows:

Neutral Geometry Axiom System (NGAS). Suppose *point* and *line* are undefined. Each *line* that exists is a point set. Consider the following axioms:

Axiom 1. If L is a line, then there exists at least two points belonging to L .

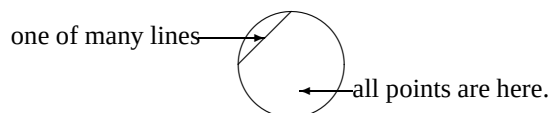
Axiom 2. If L is a line, then there exists at least one point that is not on L .

Axiom 3. There exists at least one line.

Axiom 4. If A and B are different points, then there exists at least one line that contains both A and B .

An **interpretation** of an axiom system is a set of meanings for the undefined terms. These meanings are assigned without any regard for whether the axioms become true statements about the undefined terms. We give three examples of an interpretation for NGAS:

- I-1.** By *point* we mean *Calvin*, *Hobbes*, or 1; that is, the set of points is $\{\text{Calvin}, \text{Hobbes}, 1\}$. By *line* we mean one of the following combinations: $\{\text{Calvin}, 1\}$, $\{\text{Hobbes}, 1\}$. Thus, we have given meaning to *point* and *line*, albeit a little silly, but meaningful nevertheless.
- I-2.** By *point* we mean any point on the circle or in the interior of the circle, in other words, each point on the circle or in its interior is a *point*. By *line* we mean any possible chord of the circle; that is, each chord of the circle is a *line*. Thus, there are as many *lines* as chords of the circle. Consider the diagram below:



I-3. Consider a fictitious goldfish bowl containing two goldfish in water and sitting on a table. Suppose there is one goldfish lying on the table beside the bowl. By *point*, we mean one of the goldfish and, by *line*, we mean the two goldfish in the bowl.

A **model** for an axiom system is an interpretation for the axiom system in which the axioms are true statements about the undefined terms.

Note that interpretation I-1 is not a model for NGAS even though Axioms 1, 2, 3 are true statements for I-1. Note that *Calvin* and *Hobbes* are not contained in a single line. Thus, Axiom 4 is not satisfied. However, I-2 is a model for NGAS because Axioms 1 - 4 are all true statements for the interpretation.

Problem 1. For interpretation I-1, redefine *line* so that all four axioms are satisfied.

Problem 2. Is I-3 a model? Explain.

Optional Material

Now, consider the following axiom systems:

POS (Partially Ordered Space) The words *point* and *precedes* are undefined. We have the following axioms:

Axiom 1. If the point x precedes the point y , then x is different from y .

Axiom 2. If each of x , y , and z is a point such that x precedes y and y precedes z , then x precedes z .

Problem POS. Give a model for the axiom system POS.

TOS (Totally Ordered Space) The words *point* and *precedes* are undefined. We have the following axioms:

Axiom 1. If each of x and y is a point, then either x precedes y or y precedes x .

Axiom 2. If the point x precedes y , then x is different from y .

Axiom 3. If each of x , y , and z is a point such that x precedes y and y precedes z , then x precedes z .

Problem TOS. Give a model for the axiom system TOS.

Problem TOS-TH. Create one theorem for this system.

Problem X. Give a model for Axiom System POS that is not a model for Axiom System TOS.

BAS (Boolean Algebra System) Suppose 0 and 1 are undefined with symbols $+$ and $\times$ satisfying the following axioms:

Axiom $+$.

+	0	1
0	0	1
1	1	1

Axiom $\times$.

$\times$	0	1
0	0	0
1	0	1

Problem BAS. Give a model for the Axiom System BAS.

Problem BAS-TH. Create one theorem for this system.

Chapter 2

Neutral Geometry

Neutral Geometry (sometimes called Absolute Geometry) is that geometry that is common to both Euclidean, Hyperbolic, and Elliptical geometries.

2.1 Consequences of Axioms 0, 1, 2, 3, and 4

The words *point* and *line* are undefined. By *point set*, we mean a set each element of which is a *point*.

Axiom 0. Each line is a point set.

Axiom 1. If L is a line, then there exist at least two points belonging to L .

Axiom 2. If L is a line, then there exists at least one point that does not belong to L .

Axiom 3. There exists at least one line.

Problem 1. What is the minimum number of points possible based on the Axioms 0, 1, 2, 3 above? Prove your answer.

Problem 2. State the theorem you discovered and proved in Problem 1.

Problem 3. Find a model that satisfies Axioms 0, 1, and 3 but not Axiom 2.

Problem 4. Create a model for Axioms 0, 1, 2, and 3 in which at least two points belong to no line.

Problem 5. Prove or disprove the following:

Each point belongs to a line.

I think you would agree that any geometry that contains two points not belonging to any line is not interesting. Therefore, we need the next axiom.

Axiom 4. If A and B are different points, then there exists at least one line that contains both A and B .

Problem 6. Find the invalid line of the following “proof.”

Proof. By Axiom 3, there is some line L . By Axiom 1, L contains two points A and B . Thus, for each point A , there is a line containing A . $\square$

Problem 7. State and prove the theorem that we “attempted” to prove in Problem 6.

Problem 8. Is it possible that some point belongs to only one line? Prove your response.

Problem 9. Find a model that satisfies our axiom system in which there are two different points A and B such that at least two different lines each contain both A and B .

We can see from Problem 9 that our axiom system is lacking. Thus, we improve our previous Axiom 4 to the following:

Axiom 4. If A and B are different points, then there exists exactly one line that contains both A and B .

Now, because of our new and improved version of Axiom 4, for the first time, we can set forth the following notation:

Notation 1. By the symbols $\overleftrightarrow{AB}$, we mean that A and B are two different points and $\overleftrightarrow{AB}$ denotes *the* line containing A and B .

Definition 1. The statement that points are collinear means that there exists some line containing each of the points.

Problem 10. What would we mean by the statement that points are non-collinear?

Problem 11. Is it true that given a point that there exist two more points such that the three points are collinear? Prove your answer.

Problem 12. Can we say each point is not in some line? Prove your answer.

Problem 13. Suppose L and H are different lines. What can you say about the number of points that can belong to both L and H ? Prove your answer.

Problem 14. Define plane.

Problem 15. Create and prove a theorem concerning planes.

Do not proceed further until the class is finished with this section.

Summary of pages 9 through 10.

Point and *line* are undefined. A *point set* is a set each element of which is a *point*.

Axiom 0. Each line is a point set.

Axiom 1. If L is a line, then there exist at least two points belonging to L .

Axiom 2. If L is a line, then there exists at least one point that is not on L .

Axiom 3. There exists at least one line.

Axiom 4. If A and B are different points, then there exists only one line that contains both A and B .

Notation 1. By the symbols $\overleftrightarrow{AB}$, we mean that A and B are two different points and $\overleftrightarrow{AB}$ denotes the line containing A and B .

Theorem 1. There exist at least three different points.

Theorem 2. If P is a point, then there is a line containing P .

Theorem 3. If P is a point, then there exist two different lines containing P .

Definition 1. The statement that points are collinear means that there exist some line containing each of the points. Points are said to be non-collinear if and only if no line contains each of the points.

Theorem 4. If P is a point, then there are two different points Q and R such that P , Q , and R are non-collinear.

Theorem 5. If P is a point, then there exists at least one line that does not contain P .

Theorem 6. If L and H are two different lines, then there is at most one point belonging to both L and H .

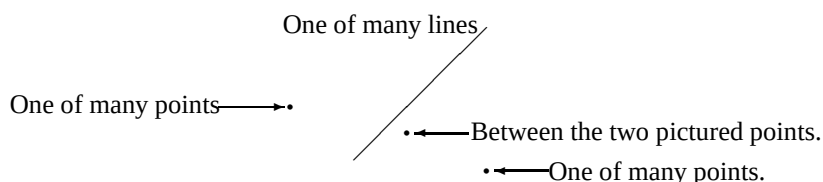
2.2 Consequences of Axioms 0, 1, 2, 3, 4, 5, 6, and 7

We now have an additional undefined term to add to *point* and *line*. That word is *between* and satisfies Axioms 5 - 8.

Notation 2. ABC means that each of A , B , and C is a point such that B is *between* A and C .

Before proceeding, we must develop a motivation for the axioms that follow. Suppose we say nothing about *between* other than it is an undefined term. Consider the following model.

Example 7. Suppose by *point* we mean point in the plane, by *line* we mean the normal “straight” line of the plane and by B is *between* A and C we mean that the point A is to the left of the point B and the point C is to the right of B . See figure below:



Note that it is possible for a point to be between two other points but the three points not be collinear. This we do not want. Thus, we must have the following axiom.

Axiom 5. If ABC , then A , B , and C are three different points of some line and CBA .

Problem 16. If XYZ , then can we say X , Y , Z are collinear? Explain.

Problem 17. Create a model for our axiom system that violates the next axiom.

Axiom 6. If A , B , and C are three different points of a line, then only one of the following holds: ABC , BCA , CAB .

Problem 18. Suppose ABC and ACD . Is it possible that two of A , B , C , and D are the same? Do they all belong to the same line or is it possible that A , B , and C belong to one line and A , C , and D belong to a second line?

Definition 2. $ABCD$ means that ABC , ABD , ACD , and BCD .

Axiom 7. If A , B , C and D are four different collinear points and ABC , then only one of the following is true: $ABCD$, $ABDC$, $ADBC$, $DABC$.

Problem 19. Suppose ABC and ACD . Prove that ABD and BCD .

Problem 20. Suppose ABD and BCD . Prove that ABC and ACD . Hint: Instead of repeating a proof of Problem 19, attempt to apply Problem 19.

One of the following two conjectures is true (a theorem) and one is false. Determine which is which. Prove the theorem and find a counter example for the other and fix that one so that a theorem results.

Conjecture 21. If ABC and BCD , then $ABCD$.

Conjecture 22. If ABD and ACD , then ABC or ACB .

Problem 23. Suppose ABC and ABD and $C \neq D$. Prove the following:

- (a) A, B, C , and D are four distinct, collinear points.
- (b) BCD or BDC .
- (c) ACD or ADC .

Problem 24. (Group work allowed) Suppose L is a line and A is a point of L . Using the notion of *between*, define two subsets, S and T of L such that

- (a) each point of L is in S or in T ,
- (b) A is the only point in both S and T ,
- (c) A is between each point of S and each point of T that is different from A , and
- (d) A is not between two points of S and is not between two points of T .

Do not proceed further until the class is finished with this section.

Summary of pages 12 and 13

The words *point* and *line* are undefined in our geometry. The new word *between* is undefined and satisfies the axioms below.

Notation 2. ABC means that A , B , and C are points such that B is between A and C .

Axiom 5. If ABC , then A , B , and C are three different points of some line and CBA .

Axiom 6. If A , B , and C are three different points of a line, then only one of the following holds: ABC , BCA , CAB .

Theorem 7. If ABC and ACD , then A , B , C , and D are four different points and are collinear.

Definition 2. $ABCD$ means that ABC , ABD , ACD , and BCD .

Axiom 7. If A , B , C , and D are four different collinear points and ABC , then only one of the following is true: $ABCD$, $ABDC$, $ADBC$, $DABC$.

Theorem 8. If ABC and ACD , then ABD and BCD .

Theorem 9. If ABD and BCD , then ABC and ACD .

Theorem 10. If ABC and BCD , then $ABCD$.

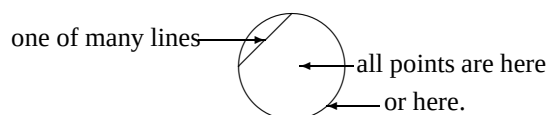
Theorem 11. If ABD and ACD and $B \neq C$, then ABC or ACB .

Theorem 12. If ABC and ABD and $C \neq D$, then BCD or BDC .

Theorem 13. If ABC and ABD and $C \neq D$, then ACD or ADC .

2.3 Consequences of Axioms 0, 1, 2, 3, 4, 5, 6, 7, and 8

Example 8. Remember the model where *point* means a point on the circle or interior to the circle and *line* means a chord of the circle. *Between* is understood in the normal sense of the word. See below.



This is a model for Axioms 1 - 7; however, the following axiom is not satisfied. In our geometry, we definitely want each pair of points to be followed by another point and, for each pair of points, there must be a point between them as stated in the following axiom.

Axiom 8. If A and B are different points, then

- (a) there is a point C such that ABC ,
- (b) there is a point D such that ADB ,
- (c) there is a point E such that EAB .

Problem 25. Modify the model of the above example so that Axiom 8 is true.

Definition 3. Suppose A and B are two different points. Then AB (called “the interval AB ”) is the set to which X belongs if and only if $X = A$, AXB , or $X = B$; that is, AB is the set of all points between A and B inclusive. Moreover, A and B are called the end points of AB .

Problem 26. Do all intervals contain at least three points?

Problem 27. For each pair of points, A and B , prove that $AB = BA$.

Problem 28. Prove that every point in the interval AB is also a member of $\overleftrightarrow{AB}$.

Problem 29. Suppose AB is an interval and APB . Prove the following:

- (a) AP is a subset of AB , and PB is a subset of AB .
- (b) $AP \cup PB$ is a subset of AB .
- (c) if x is in AB , then x is in AP or x is in PB .
- (d) AB is a subset of $AP \cup PB$.
- (e) $AB = AP \cup PB$.

Problem 30. Suppose AB is an interval and APB . Prove that $AP \cap PB = \{P\}$.

Problem 31. Suppose each of AB and CD is an interval where $AB = CD$. Prove that $A = C$ or $A = D$. What else can you prove?

Problem 32. Find a model satisfying all axioms stated thus far where there exists three different points A , B , and C , a line L not containing A , B , and C , contains a point of AB but does not contain a point of AC or BC .

Do not proceed further until the class is finished with this section.

Summary of pages 15 and 15

Axiom 8. If A and B are different points, then

- (a) there is a point C such that ABC ,
- (b) there is a point D such that ADB ,
- (c) there is a point E such that EAB .

Definition 3. Suppose A and B are two different points. Then AB (called “the interval AB ”) is the set to which X belongs if and only if $X = A$, AXB , or $X = B$; that is, AB is the set of all points between A and B inclusive. Moreover, A and B are called the end points of AB .

Theorem 14. If AB is an interval, then there exists a point in AB that is neither A nor B .

Theorem 15. If A and B are different points, then $AB = BA$.

Theorem 16. If A and B are different points, then every point in the interval AB is a member of $\overleftrightarrow{AB}$; that is, AB is a subset of $\overleftrightarrow{AB}$.

Theorem 17. If AB is an interval and APB , then $AB = AP \cup PB$.

Theorem 18. If AB is an interval and APB , then $AP \cap PB = \{P\}$.

Theorem 19. If each of AB and CD is an interval and $AB = CD$, then $A = C$ and $B = D$ or $A = D$ and $B = C$.

2.4 Consequences of Axioms 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9: Part I

Axiom 9. (Pasch) If A, B, C are three distinct non-collinear points and L is a line that does not contain A, B , or C but does contain a point of AB , then L contains a point of AC or BC but not both.

Problem 33. Suppose L is a line. Prove there is a set U containing at least two points such that no point of L is between each pair of points in U .

Definition 4. Suppose L is a line. Then the statement that M is a side of L means that there is a point O not on L such that M is the set to which the point X belongs if and only if X is O or the interval XO does not contain a point of L . M is called “ O -side of L .”

Problem 34. Suppose L is a line and Q is a point not on L . Complete: The statement that a point X does not belong to the Q -side of L means

Problem 35. Suppose L is a line, A and A' are distinct points not on L , and no point of L is between A and A' . Prove that if X is a point on the A -side of L , then X is on the A' -side of L . Explain why the the A -side of L is the A' -side of L .

Problem 36. Suppose L is a line. Prove the following:

- (a) L has a side.
- (b) L has a second side in addition to the one in (a).
- (c) The two sides of (a) and (b) have no point in common.
- (d) There is not a third side.

Definition 5. The statement that the point set M is convex means that for each pair of distinct points P and Q of M , the interval PQ is contained in M .

Problem 37. Prove that the following are convex: each line, each interval, each side of each line. Prove in any order you wish and see if you can use one fact to prove the other.

Problem 38. Prove that the intersection (common part) of a convex set with a convex set is a convex set.

Do not proceed further until the class is finished with this section.

Summary of page 17

Axiom 9. (Pasch) If A, B, C are three distinct non-collinear points and L is a line that does not contain A, B , or C but does contain a point of AB , then L contains a point of AC or BC but not both.

Definition 4. Suppose L is a line. Then the statement that M is a side of L means that there is a point O not on L such that M is the set to which the point X belongs if and only if X is O or the interval XO does not contain a point of L . M is called “ O -side of L .”

Theorem 20. If L is a line, A and A' are distinct points not on L , and no point of L is between A and A' , then the A -side of L is the A' -side of L .

Theorem 21. If L is a line, then L has exactly two sides, the two sides have no point in common and neither contains a point of L .

Definition 5. The statement that the point set M is convex means that for each pair of distinct points P and Q of M the interval PQ is contained in M .

Theorem 22. All lines, intervals and sides of lines are convex.

Theorem 23. The intersection (common part) of a convex set with a convex set is a convex set.

2.5 Consequences of Axioms 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9: Part II

Problem 39. Suppose A, B, C are three distinct non-collinear points. Prove there exists a point that belongs to the C -side of $\overleftrightarrow{AB}$, the A -side of $\overleftrightarrow{BC}$, and the B -side of $\overleftrightarrow{AC}$.

Definition 6. Suppose A, B, C are three distinct non-collinear points. Then $\triangle ABC$ (called triangle ABC) is the set $AB \cup BC \cup AC$.

Definition 7. Suppose $\triangle ABC$ is a triangle. Then

- (a) the statement that a point is a vertex of $\triangle ABC$ means that point is A, B or C , and
- (b) the statement that an interval is a side of $\triangle ABC$ means that interval is AB, BC or AC .

Definition 8. Suppose A and B are different points. Then the statement that $\overrightarrow{AB}$ is a ray (read “ray AB ”) means that AB is the set of all points X such that $X = A, AXB, X = B$, or ABX .

Definition 9. If A, B, C are three distinct non-collinear points, then $\angle ABC$ (called angle ABC) is the set $\overrightarrow{BA} \cup \overrightarrow{BC}$.

Definition 10. If A, B, C are three distinct non-collinear points, then

- (a) the interior of $\angle ABC$ is the intersection (common part) of the A -side of BC and the C -side of BA and
- (b) the interior of $\triangle ABC$ is the intersection (common part) of the interior of $\angle ABC, \angle ACB$, and $\angle BAC$.

Problem 40. Prove that the interior of each angle and each triangle is convex.

Definition 11. The side of a triangle that does not contain a given vertex is called the side opposite that vertex.

Problem 41. Prove that each line containing a vertex of a triangle and a point in the interior of the triangle contains a point of the side opposite that vertex.

Problem 42. Define “quadrilateral.”

Problem 43. Conjecture a theorem concerning quadrilaterals and prove the theorem.

Do not proceed further until the class is finished with this section.

Summary of page 19

Theorem 24. Suppose A, B, C are three distinct non-collinear points. Then there exists a point that belongs to the C -side of $\overleftrightarrow{AB}$, the A -side of $\overleftrightarrow{BC}$, and the B -side of $\overleftrightarrow{AC}$.

Definition 6. Suppose A, B, C are three distinct non-collinear points. Then $\triangle ABC$ (called triangle ABC) is the set $AB \cup BC \cup AC$.

Definition 7. Suppose $\triangle ABC$ is a triangle. Then

- (a) the statement that a point is a vertex of $\triangle ABC$ means that point is A, B or C , and
- (b) the statement that an interval is a side of $\triangle ABC$ means that interval is AB, BC or AC .

Definition 8. Suppose A and B are different points. Then the statement that $\overrightarrow{AB}$ is a ray (read “ray AB ”) means that AB is the set of all points X such that $X = A, AXB, X = B$, or ABX .

Definition 9. If A, B, C are three distinct non-collinear points, then $\angle ABC$ (called angle ABC) is the set $\overrightarrow{BA} \cup \overrightarrow{BC}$.

Definition 10. If A, B, C are three distinct non-collinear points, then

- (a) the interior of $\angle ABC$ is the intersection (common part) of the A -side of BC and the C -side of BA and
- (b) the interior of $\triangle ABC$ is the intersection (common part) of the interior of $\angle ABC, \angle ACB$, and $\angle BAC$.

Theorem 25. The interior of each angle and each triangle is convex.

Definition 11. The side of a triangle that does not contain a given vertex is called the side opposite that vertex.

Theorem 26. Each line containing a vertex of a triangle and a point in the interior of the triangle contains a point of the side opposite that vertex.

2.6 Future Study

For those who would like to continue this study consult the following:

- *Euclidean and Non-Euclidean Geometries* by Marvin Jay Greenberg
- *Monographs on topics of Modern Mathematics* edited by J. W. A. Young

In addition to the excellent sources above for study of Non-Euclidean Geometry, consult the following :

- *Non-Euclidean Geometry* by Herbert Meschkowski
- *Non-Euclidean Geometry* by Roberto Bonola and H.C. Carlsaw

Language and Proof (LP) Worksheets 1 - 4

LP Worksheet 1

Remember you can use any information that appears on page 5 prior to Theorem 3 and all pages prior to page 5.

Problem 1. (Theorem 3) Suppose $A \subseteq B$ and $B \subseteq C$. Prove that $A \subseteq C$.

LP Worksheet 2

Use may use any information prior to Theorem 4 on page 5.

Problem 2. Suppose each of A and B is a set. Prove $A \subseteq A \cup B$.

Problem 3. Suppose each of A and B is a set. Prove $A \cap B \subseteq A$.

Problem 4. Suppose each of A and B is a set such that $A \subseteq B$. Prove $A \cup B = B$.

Problem 5. Suppose each of A and B is a set such that $A \cup B = B$. Prove $A \subseteq B$.

LP Worksheet 3

Use may use any information prior to Theorem 6 on page 5.

Problem 6. Suppose U is a set and each of A and B is a subset of U . Complements are taken with respect to U . Give an example of a diagram as evidence that $(A \cup B)^c = A^c \cap B^c$.

Problem 7. Suppose U is a set and each of A and B is a subset of U . Complements are taken with respect to U . Give an example of a diagram as evidence that $(A \cap B)^c = A^c \cup B^c$.

Problem 8. Suppose U is a set and each of A and B is a subset of U . Complements are taken with respect to U . Prove either $(A \cup B)^c = A^c \cap B^c$ or $(A \cap B)^c = A^c \cup B^c$.

LP Worksheet 4

Problem 9. By *point* we mean *Calvin*, *Hobbes*, or 1; that is, the set of points is $\{\textit{Calvin}, \textit{Hobbes}, 1\}$. By *line*, we mean one of the following combinations: $\{\textit{Calvin}, 1\}$, $\{\textit{Hobbes}, 1\}$. Thus, we have given meaning to *point* and *line*. Do we have a model for NGAS? If not, explain which axioms fail to be satisfied.

Problem 10. Consider a fictitious goldfish bowl containing two goldfish in water and sitting on a table. Suppose there is one goldfish lying on the table beside the bowl. By *point* we mean one of the goldfish and by *line* we mean the two goldfish in the bowl. Do we have a model for NGAS? If not, explain which axioms fail to be satisfied.

Neutral Geometry (NG) Worksheets 1 - 18

NG Worksheet 1

Problem 1. What is the minimum number of points possible based on the Axioms 0, 1, 2, and 3 above? Prove your answer.

Problem 2. State the theorem you discovered and proved in Problem 1.

Problem 3. Find a model that satisfies Axioms 0, 1, and 3 but not Axiom 2.

(a) *point* means

(b) *line* means

Explanation

NG Worksheet 2

Problem 4. Create a model for Axioms 0,1, 2, and 3 in which there are at least two points x and y with the property that no line contains both x and y .

point means

line means

Explanation.

Problem 5. Is the following a theorem of the axiom system?

Each point belongs to some line.

Yes ____ No ____

Prove your response.

NG Worksheet 3

Problem 6. Find the invalid line of the proof.

Proof. By Axiom 3, there is some line L . By Axiom 1, L contains two points A and B . Thus, for each point A , there is a line containing A . $\square$

Explanation.

Problem 7. State and prove the theorem that we attempted to prove in Problem 6.

Statement.

Proof.

NG Worksheet 4

Problem 8. Is it possible that some point belongs to just one line and not two lines? Prove your response.

NG Worksheet 5

Problem 9. Find a model that satisfies our axiom system in which there are two different points A and B such that two different lines each contain both A and B . Prove your response.

NG Worksheet 6

Problem 10. What we would mean by the statement that points are not collinear?

Problem 11. Is it true that given a point that there exist two more points such that the three points are non-collinear? Prove your answer.

Problem 12. Can we say that given a point that there is a line that does not contain the point? Prove your answer.

Problem 13. Suppose L and H are two different lines. What can you say about the points that belong to both L and H ? Prove your answer.

NG Worksheet 7

Problem 16. Based on our axiom system, if XYZ , can we say X, Y, Z are collinear? Explain.

Problem 17. Create a model for our axiom system that violates the following axiom.

Axiom 6. If A, B , and C are three different points of a line, then only one of the following holds: ABC, BCA, CAB .

Problem 18. Suppose ABC and ACD . Is it possible that two of A, B, C , and D are the same? Do they all belong to the same line or is it possible that A, B , and C belong to one line and A, C , and D belong to a second line?

NG Worksheet 8

Problem 19. Suppose ABC and ACD . Prove ABD and BCD .

Problem 20. Suppose ABD and BCD . Prove ABC and ACD . Instead of repeating a proof of Problem 19, attempt to prove Problem 20 by applying Problem 19.

NG Worksheet 9

One of the following two conjectures is true (a theorem) and one is false. Determine which is which. Prove the theorem and find a counter example for the other.

Conjecture 21. If ABC and BCD , then $ABCD$.

Conjecture 22. If ABD and ACD , then ABC or ACB .

NG Worksheet 10

Problem 23. Suppose ABC and ABD and $C \neq D$. Prove the following:

(a) A, B, C , and D are four distinct, collinear points.

(b) BCD or BDC .

(c) ACD or ADC .

NG Worksheet 11

Problem 24. (Group work allowed) Suppose L is a line and A is a point of L . Using the notion of *between*, define two subsets, S and T of L such that

- (a) each point of L is in S or in T ,
- (b) A is the only point in both S and T ,
- (c) A is between any other point of S and any other point of T and
- (d) A is not between each pair of points in S and between each pair of points in T .

NG Worksheet 12

Problem 25. Modify the model of Example 8 on page 15 so that Axiom 8 is true.

Problem 26. Do all intervals contain at least three points? Prove your response.

Problem 27. For each pair of points, A and B , prove that $AB = BA$.

NG Worksheet 13

Problem 28. Prove that each point in the interval AB is a member of $\overleftrightarrow{AB}$; that is, AB is a subset of $\overleftrightarrow{AB}$.

Problem 29. Suppose AB is an interval and APB . Prove

- (a) AP is a subset of AB , and PB is a subset of AB ,
- (b) $AP \cup PB$ is a subset of AB ,
- (c) if x is in AB , then x is in AP or x is in PB ,
- (d) AB is a subset of $AP \cup PB$, and
- (e) $AB = AP \cup PB$.

NG Worksheet 14

Problem 30. Suppose AB is an interval and APB . Prove $AP \cap PB = \{P\}$.

Problem 31. Suppose each of AB and CD is an interval where $AB = CD$. Prove $A = C$ or $A = D$. What else can you prove?

Problem 32. Find a model satisfying all axioms stated thus far where there are three different points A , B , and C and a line L that does not contain A , B , and C , contains a point of AB but no point of AC or BC .

NG Worksheet 15

Problem 33. Suppose L is a line. Find a set U containing at least two points such that no point of L is between each pair of points in U .

Problem 34. Suppose L is a line and Q is a point not on L . Complete: The statement that X does not belong to the Q -side of L means

Problem 35. Suppose L is a line, A and A' are distinct points not on L , and no point of L is between A and A' . Prove that if X is a point on the A -side of L , then X is on the A' -side of L . Explain why the A -side of L is the A' -side of L .

Problem 36. Suppose L is a line. Prove the following:

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- Charles A. Coppin *Journal of Inquiry-Based Learning in Mathematics*

NG Worksheet 17

Problem 37. Prove that the following are convex:

(a) Each line.

(b) Each interval.

(c) Each side of each line.

NG Worksheet 18

Problem 38. Prove that the intersection (common part) of a convex set with a convex set is a convex set.

Problem 39. Suppose A, B, C are three distinct non-collinear points. Prove there exists a point that belongs to the C -side of $\overleftrightarrow{AB}$, the A -side of $\overleftrightarrow{BC}$, and the B -side of $\overleftrightarrow{AC}$.